

Teacher Ratings — Probability & Hypothesis Testing

Dataset: TeachingRatings.csv

This notebook answers three questions:

1. $P(\text{evaluation score} > 4.5)$
2. $P(3.5 < \text{evaluation score} < 4.2)$
3. A two-tailed z-test comparing a regional basketball league's scoring average to a historic pro mean

For Q1 and Q2, the assignment doesn't specify whether to use the **actual data** (empirical probability) or a **fitted normal distribution** (theoretical probability). Both are computed below so you can pick whichever your course expects — check your assignment sheet or lecture notes if unsure, since the two approaches give noticeably different answers for Q2.

Setup and data load

```
import pandas as pd
import numpy as np
from scipy import stats

df = pd.read_csv("TeachingRatings.csv")
print("Shape:", df.shape)
df.head()
```

Shape: (463, 13)

	rownames	minority	age	gender	credits	beauty	eval	division	native	tenure	students	allstudents	prof
0	1	yes	36	female	more	0.289916	4.3	upper	yes	yes	24	43	1
1	2	no	59	male	more	-0.737732	4.5	upper	yes	yes	17	20	2
2	3	no	51	male	more	-0.571984	3.7	upper	yes	yes	55	55	3
3	4	no	40	female	more	-0.677963	4.3	upper	yes	yes	40	46	4
4	5	no	31	female	more	1.509794	4.4	upper	yes	yes	42	48	5

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
eval_scores = df["eval"]

n = len(eval_scores)
mean = eval_scores.mean()
std = eval_scores.std(ddof=1) # sample standard deviation
```

```
print(f"n      = {n}")
print(f"mean = {mean:.4f}")
print(f"std  = {std:.4f}")
```

```
n      = 463
mean   = 3.9983
std    = 0.5549
```

✓ Q1: $P(\text{evaluation score} > 4.5)$

Empirical: proportion of rows in the dataset with `eval > 4.5`.

Normal model: treat `eval ~ Normal(mean, std)` using the sample mean/std above, compute the z-score for 4.5, and use the standard normal survival function.

```
# Empirical probability
p_emp_q1 = (eval_scores > 4.5).mean()

# Normal-model probability
z_q1 = (4.5 - mean) / std
p_norm_q1 = 1 - stats.norm.cdf(4.5, mean, std)

print(f"Empirical P(eval > 4.5)    = {p_emp_q1:.4f}  ({p_emp_q1*100:.1f}%)")
print(f"z-score for 4.5           = {z_q1:.4f}")
print(f"Normal-model P(eval > 4.5) = {p_norm_q1:.4f}  ({p_norm_q1*100:.1f}%)")
```

```
Empirical P(eval > 4.5)    = 0.1706   (17.1%)
z-score for 4.5           = 0.9042
Normal-model P(eval > 4.5) = 0.1829   (18.3%)
```

✓ Q2: $P(3.5 < \text{evaluation score} < 4.2)$

Same two approaches as Q1.

```
# Empirical probability
p_emp_q2 = ((eval_scores > 3.5) & (eval_scores < 4.2)).mean()

# Normal-model probability
z_35 = (3.5 - mean) / std
z_42 = (4.2 - mean) / std
p_norm_q2 = stats.norm.cdf(4.2, mean, std) - stats.norm.cdf(3.5, mean, std)

print(f"Empirical P(3.5 < eval < 4.2) = {p_emp_q2:.4f}  ({p_emp_q2*100:.1f}%)")
print(f"z-scores: 3.5 -> {z_35:.4f}, 4.2 -> {z_42:.4f}")
print(f"Normal-model P(3.5 < eval < 4.2) = {p_norm_q2:.4f}  ({p_norm_q2*100:.1f}%)")
```

```
Empirical P(3.5 < eval < 4.2)    = 0.3542   (35.4%)
z-scores: 3.5 -> -0.8980, 4.2 -> 0.3636
Normal-model P(3.5 < eval < 4.2) = 0.4573   (45.7%)
```

Note: the empirical and normal-model answers diverge more here (~10 points) than in Q1. The `eval` distribution is left-skewed — scores bunch up between 4.0 and 4.3 with a longer tail toward low scores — so a symmetric normal curve overstates probability mass in this middle range. If the assignment wants the real data-driven answer, use the empirical figure; if it wants the theoretical normal-distribution answer, use the normal-model figure.

✓ Q3: Two-tailed z-test — regional players vs. historic pro mean

Setup

- Historic pro mean: $\mu_0 = 12$ points/game
- Historic pro standard deviation: $\sigma = 5.5$ (treated as the known population σ for this test)
- Regional sample: $n = 36$ players, sample mean $\bar{x} = 10.7$ points/game

Hypotheses

- $H_0: \mu = 12$ — the regional players' mean is not different from the historic mean
- $H_1: \mu \neq 12$ — the regional players' mean is different from the historic mean

This is a **z-test** (not a t-test) because σ is given as a known value rather than estimated from the sample.

```
mu0 = 12
sigma = 5.5
n_players = 36
x_bar = 10.7
alpha = 0.05

se = sigma / np.sqrt(n_players)
z_stat = (x_bar - mu0) / se
p_value = 2 * (1 - stats.norm.cdf(abs(z_stat)))
z_crit = stats.norm.ppf(1 - alpha/2)

print(f"Standard error      = {se:.4f}")
print(f"z statistic         = {z_stat:.4f}")
print(f"Two-tailed p-value     = {p_value:.4f}")
print(f"Critical z (a=0.05)  = +/-{z_crit:.4f}")

if abs(z_stat) > z_crit:
    print("\nDecision: |z| > z_crit -> REJECT H0")
else:
    print("\nDecision: |z| < z_crit -> FAIL TO REJECT H0")
```

```
Standard error      = 0.9167
z statistic         = -1.4182
Two-tailed p-value  = 0.1561
Critical z (α=0.05) = +/-1.9600
```

```
Decision: |z| < z_crit -> FAIL TO REJECT H0
```

Conclusion: $|z| = 1.418$ is less than the critical value of 1.96, and the p-value (0.156) is greater than $\alpha = 0.05$. So we **fail to reject H_0** .

There isn't enough evidence at the 5% significance level to conclude the regional players' average scoring differs from the historic pro mean of 12 points/game. Failing to reject H_0 doesn't prove the means are equal — it means this sample ($n=36$) doesn't provide statistically significant evidence of a difference.

Caveat worth checking with your instructor: $\sigma = 5.5$ was used here as the assumed population standard deviation for the test. If your course instead wants you to estimate the standard deviation from the regional sample itself (making this a t-test), the result could change — the problem statement doesn't give the regional sample's own standard deviation, so that option isn't computable from the data given.